

SoleMate — Architecture & Deployment Design

Project: SoleMate (footwear ecommerce platform) **Document type:** Architecture & DevOps design
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1. Introduction

1.1 Purpose

This document describes the architecture of the SoleMate ecommerce application and the design for deploying it to AWS using modern DevOps practices: Infrastructure as Code, containerization, a serverless event tier, and a fully automated CI/CD pipeline.

1.2 Scope

It covers the application architecture (frontend, API, data, events), the target AWS cloud architecture, the network/security design, the deployment process, and the CI/CD pipeline. Diagrams are provided at the system, container, deployment, and network levels.

1.3 Goals & non-functional requirements

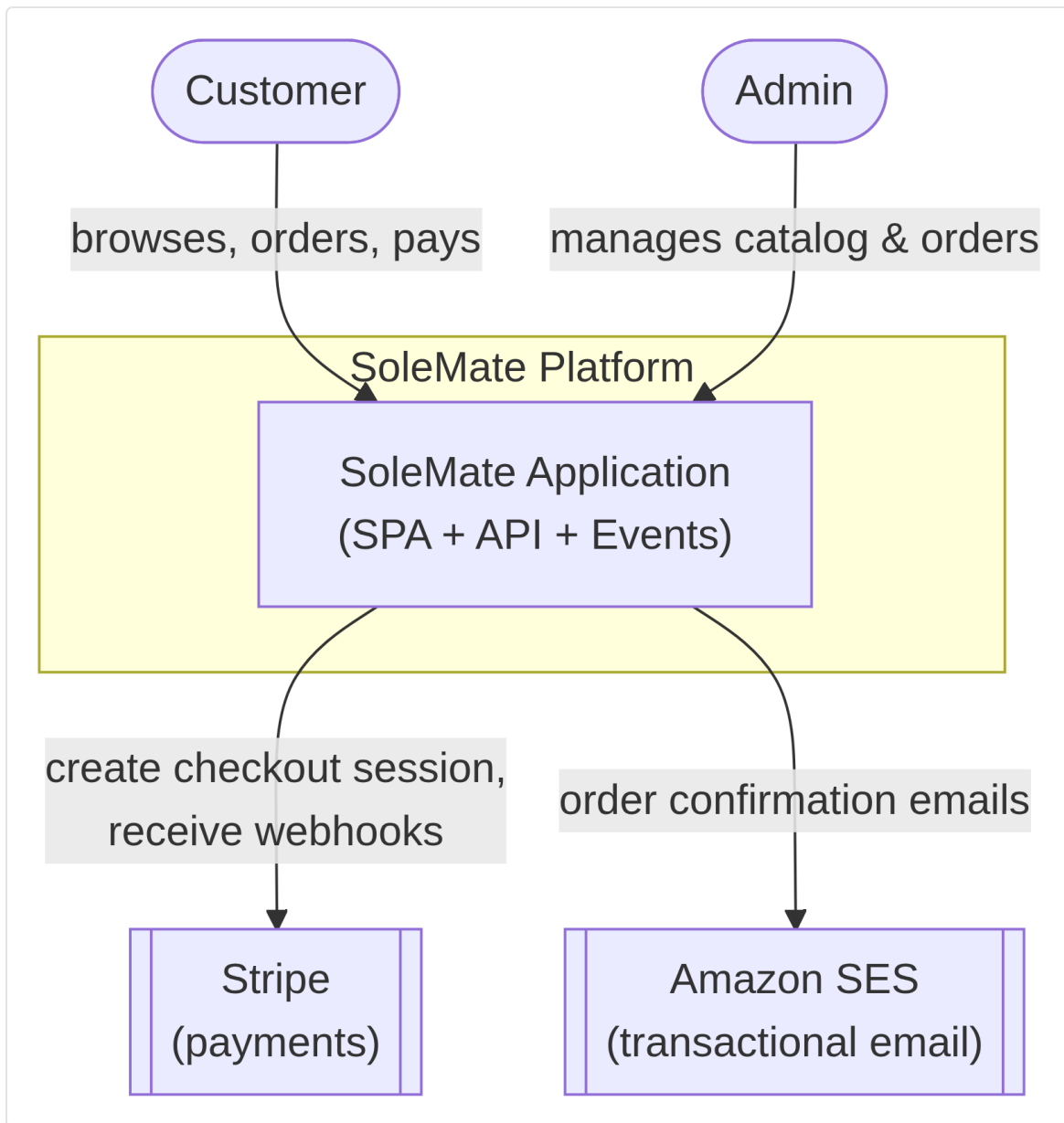
Goal	How it is addressed
Scalability	Stateless API on ECS Fargate behind an ALB with auto-scaling; serverless Lambda for spiky event work
Availability	Multi-AZ deployment of ECS tasks and RDS; managed, self-healing services
Security	Private subnets for compute/data, secrets in Secrets Manager, least-privilege IAM, HTTPS everywhere
Maintainability	100% IaC, immutable container images, one-command pipeline deploys
Observability	Centralized logs (CloudWatch), health/readiness probes, metrics & alarms
Cost-efficiency	Fargate right-sizing, serverless for bursty work, S3+CloudFront for static assets

2. Application overview

SoleMate is a classic three-tier web application.

Tier	Technology	Responsibility
Presentation	React 18 + Vite + Tailwind (SPA)	UI, cart, checkout, admin
Application	Node.js + Express REST API	Auth (JWT), products, orders, payments
Data	PostgreSQL	Users, products, orders, payments
Events	(new) AWS Lambda	Stripe webhook, image processing, email
Storage	Object storage (S3)	Product images
External	Stripe	Card payments

3. System context (C4 — Level 1)



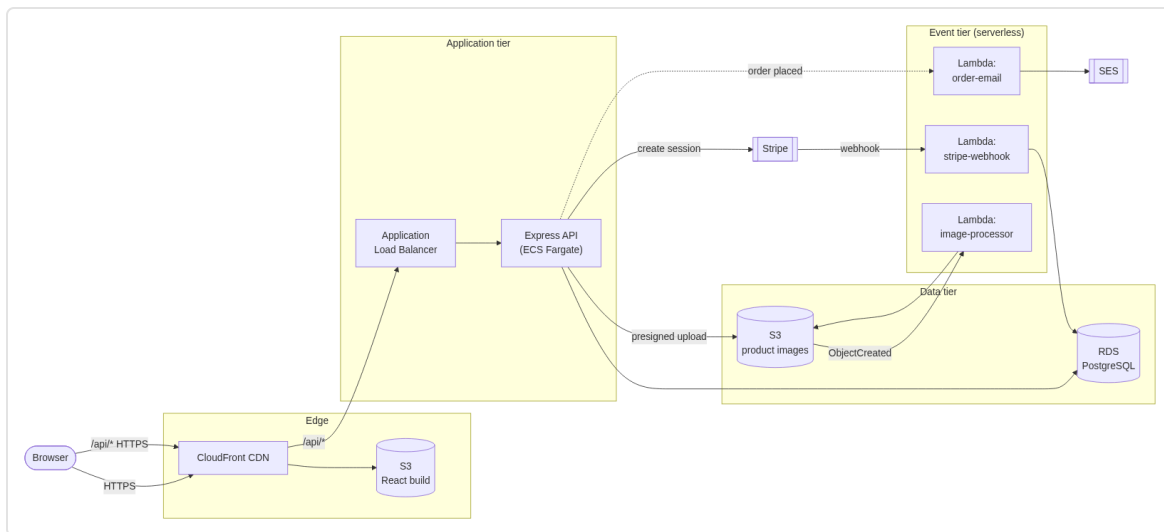
Actors

- **Customer** — browses products, manages a cart, places and pays for orders.
- **Admin** — manages the product catalog, sizes, images, and views orders.

External systems

- **Stripe** — hosted card checkout + webhooks for payment confirmation.
 - **Amazon SES** — sends order-confirmation email (replaces the old EmailJS).
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4. Container architecture (C4 — Level 2)

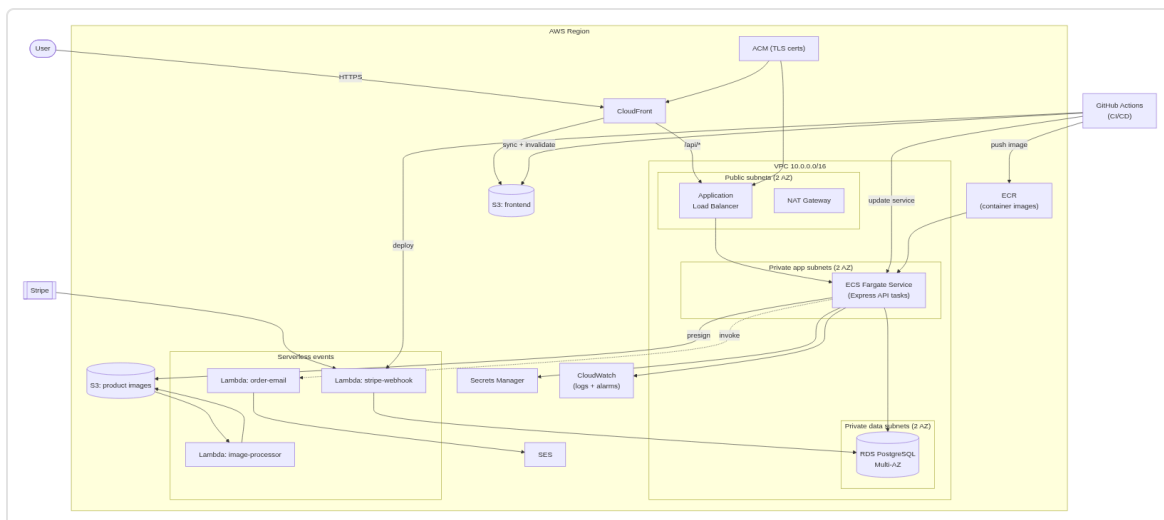


Why this split (ECS for the API, Lambda for events)?

- The Express API is a long-running request/response service with steady traffic and shared connections (a DB pool). That fits a **container on ECS Fargate**.
- The webhook, image-processing, and email tasks are **event-driven and bursty**. They map naturally to **Lambda**, which scales to zero and bills per invocation.

This is a deliberate architectural decision, not "Lambda for everything."

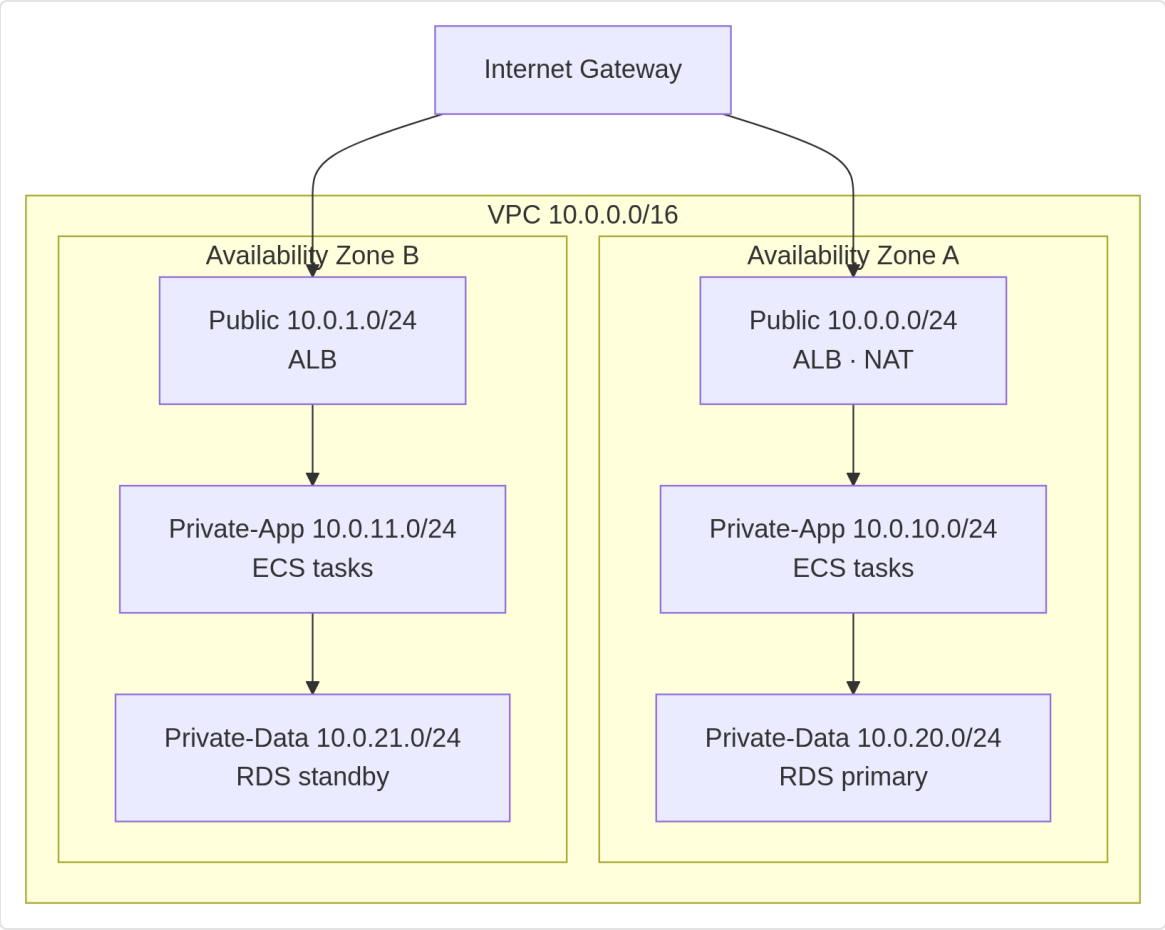
5. AWS deployment architecture



5.1 AWS service inventory

Service	Role in SoleMate
CloudFront	CDN for the SPA + TLS termination; routes <code>/api/*</code> to the ALB
S3 (frontend)	Hosts the static React build (private, served via CloudFront OAC)
S3 (images)	Stores product images; emits events to the image-processor Lambda
ALB	Load-balances and health-checks the ECS API tasks
ECS Fargate	Runs the Express API as containers (no servers to manage)
ECR	Stores versioned API container images
RDS (PostgreSQL)	Managed, Multi-AZ relational database
Lambda	Stripe webhook, image processing, order emails
Secrets Manager	DB credentials, JWT secret, Stripe keys
ACM	TLS certificates for CloudFront and the ALB
CloudWatch	Logs, metrics, dashboards, alarms
SES	Transactional email
IAM	Least-privilege roles for ECS tasks, Lambdas, and CI/CD

6. Network & security design



6.1 Subnet tiers

- **Public subnets** — only internet-facing resources (ALB, NAT Gateway).
- **Private app subnets** — ECS tasks; outbound internet via NAT only.
- **Private data subnets** — RDS; no internet route at all.

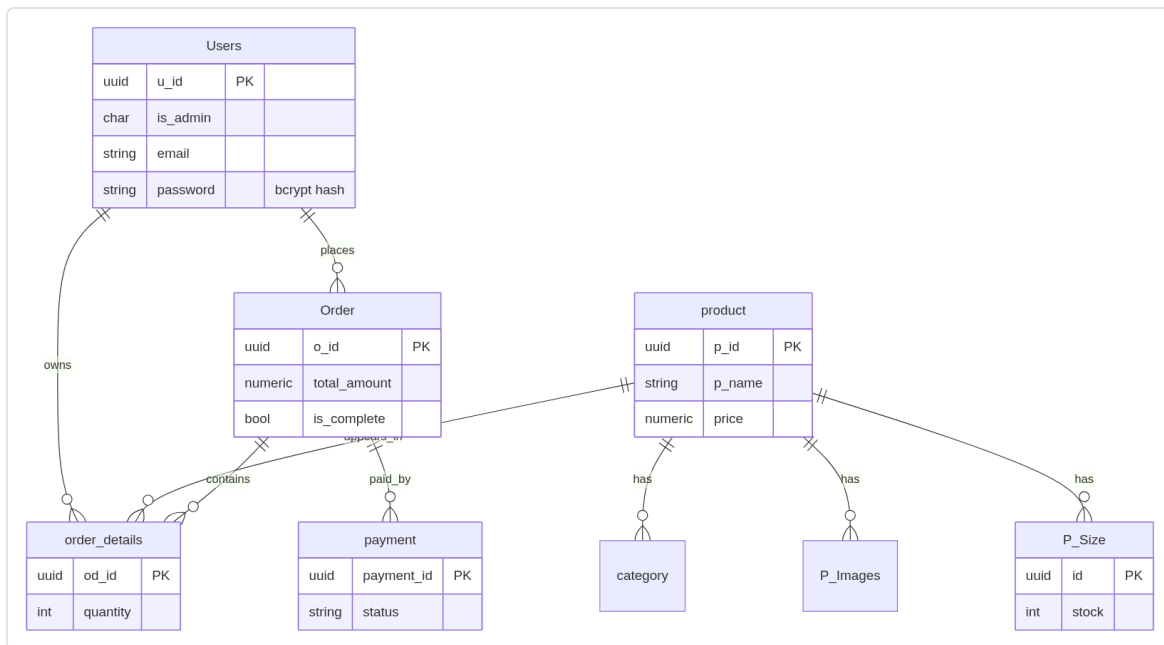
6.2 Security groups (least privilege)

SG	Inbound	From
alb-sg	443, 80	0.0.0.0/0
ecs-sg	container port (5000)	alb-sg only
rds-sg	5432	ecs-sg and the migration runner only

6.3 Secrets & identity

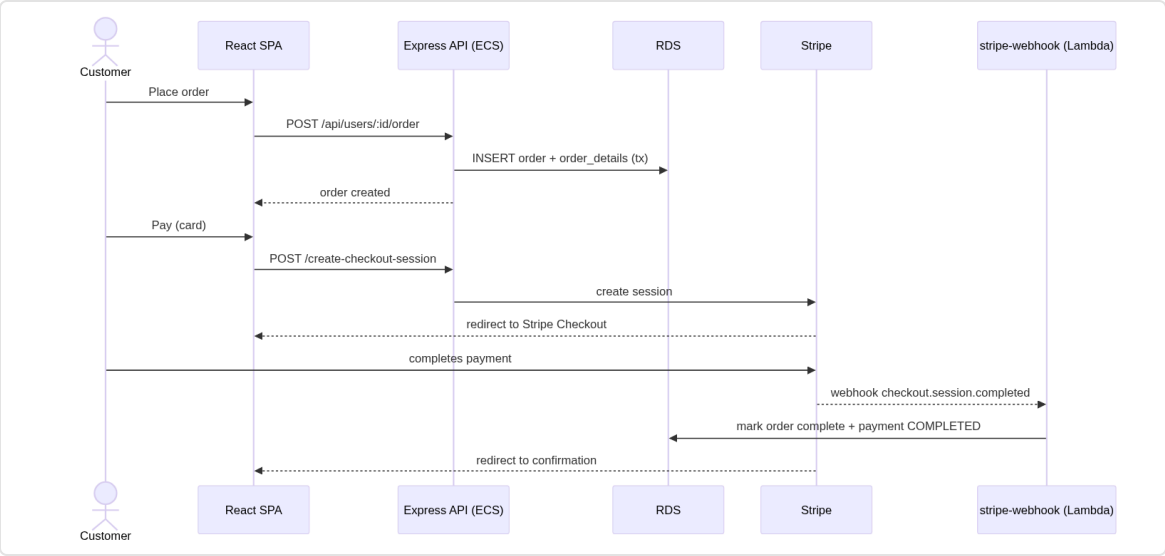
- No secrets in images or env files — all pulled from **Secrets Manager** at task start and injected into the container.
 - **IAM roles** are scoped per workload (ECS task role, each Lambda's execution role, the CI/CD deploy role) — no shared, broad credentials.
 - TLS end-to-end (CloudFront → ALB → task), HSTS via `helmet` .
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7. Data model



Full DDL: [backend/DB/schema.sql](#) . The same schema is applied to local Postgres (dev) and RDS (prod) via `npm run db:migrate` .

8. Key runtime flow — checkout



(Non-card methods — Cash on Delivery / Bank Transfer — skip Stripe and record the payment directly via the API.)

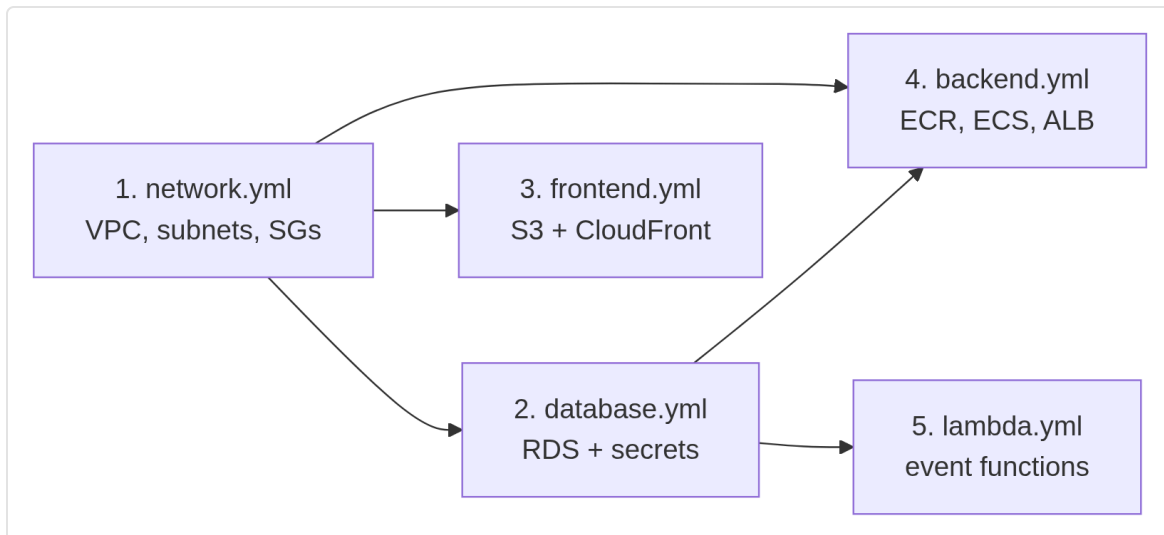
9. Deployment process

9.1 Environments

Environment	Frontend	API	Database
Local	Vite dev server	<code>npm run dev</code>	Docker Postgres
Staging	S3 + CloudFront	ECS service (1 task)	RDS (single-AZ)
Production	S3 + CloudFront	ECS service (≥2 tasks, auto-scale)	RDS Multi-AZ

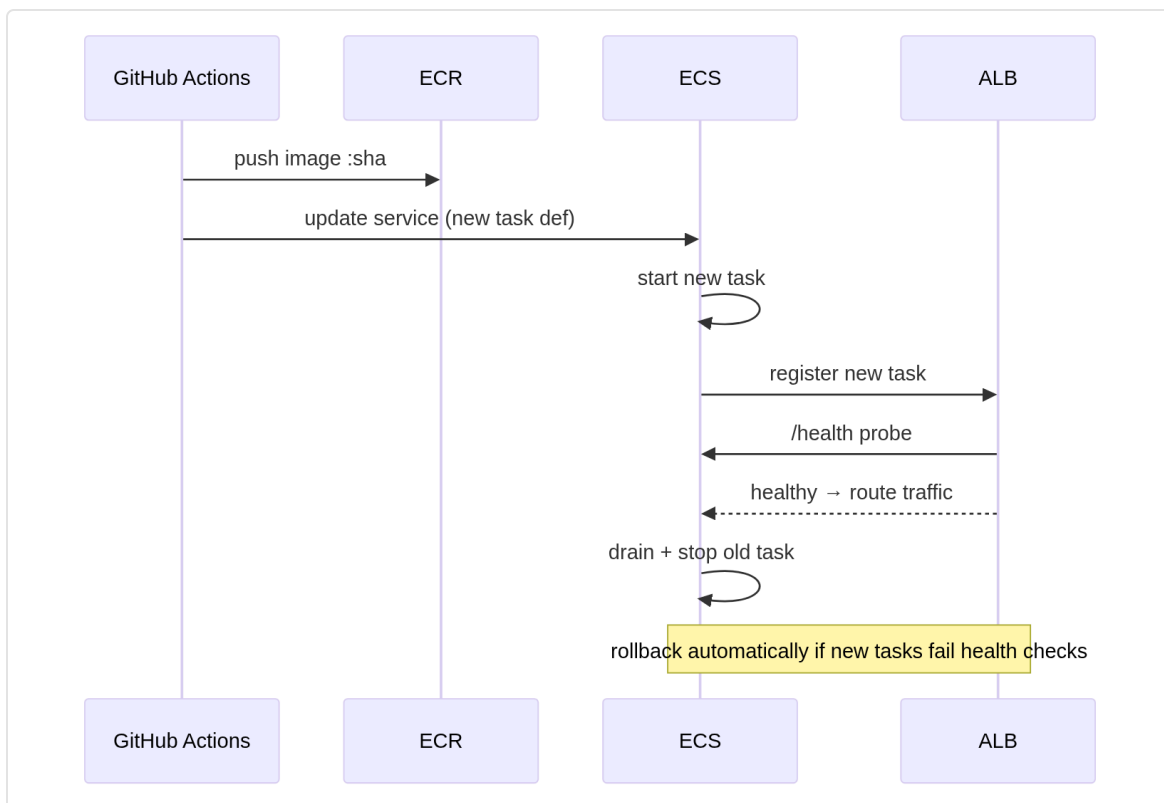
9.2 Provisioning order (CloudFormation stacks)

Stacks are deliberately split so they can be updated independently:



9.3 Rolling (zero-downtime) API deploy

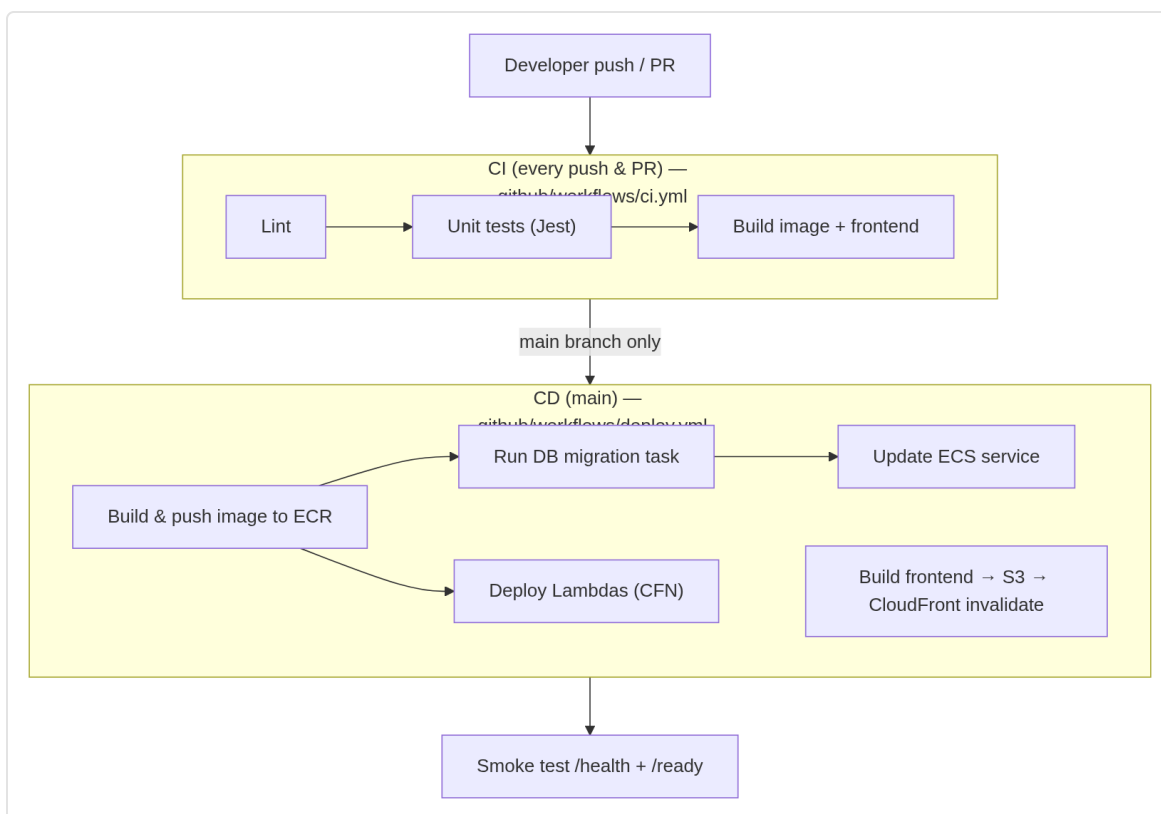
ECS performs a rolling replacement; the ALB only sends traffic to tasks that pass the `/health` check, and drains old tasks gracefully (the API handles `SIGTERM`).



9.4 Database migrations

Migrations run as a one-off **ECS task** (same image, `npm run db:migrate`) before the service update, so schema changes land before new code serves traffic.

10. CI/CD pipeline



10.1 Pipeline stages

Stage	Trigger	Action
Lint & test	every push/PR	ESLint + Jest; blocks merge on failure
Build	every push/PR	Build Docker image and frontend bundle
Push image	merge to <code>main</code>	Tag with git SHA, push to ECR
Migrate	merge to <code>main</code>	Run migration as a one-off ECS task
Deploy API	merge to <code>main</code>	Update ECS service (rolling)
Deploy Lambdas	merge to <code>main</code>	<code>aws cloudformation deploy</code> for <code>lambda.yml</code>
Deploy frontend	merge to <code>main</code>	<code>aws s3 sync</code> + CloudFront invalidation
Verify	after deploy	<code>curl /health</code> and <code>/ready</code>

10.2 Pipeline security

- AWS access uses **GitHub OIDC** (short-lived role assumption) — no long-lived AWS keys stored in GitHub.

- Image is tagged immutably with the commit SHA for traceability and rollback.
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11. Observability & operations

- **Logs:** the API logs structured JSON (`pino`) to stdout → CloudWatch Logs.
 - **Health:** `GET /health` (liveness) and `GET /ready` (DB reachable) back the ALB target-group and container health checks.
 - **Metrics & alarms:** ALB 5xx rate, ECS CPU/memory, RDS connections/CPU → CloudWatch alarms.
 - **Scaling:** ECS target-tracking on CPU (e.g. scale out at 70%).
 - **Backups:** RDS automated snapshots + point-in-time recovery.
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12. Cost considerations (illustrative)

Service	Driver	Lever
ECS Fargate	task count × size × time	right-size CPU/mem, scale to 1 task off-peak
RDS	instance class + Multi-AZ	single-AZ in staging, smallest viable class
NAT Gateway	hourly + data	one NAT (vs per-AZ) in non-prod
S3 + CloudFront	storage + egress	cache static assets aggressively
Lambda	invocations + duration	scales to zero when idle

13. Repository layout (DevOps artifacts)

```
.
├─ docs/DESIGN.md           ← this document
├─ .github/workflows/
│   ├── ci.yml              ← lint, test, build (every push/PR)
│   └─ deploy.yml           ← deploy to AWS (main)
├─ infra/cloudformation/
│   ├── network.yml         ← VPC, subnets, NAT, security groups
│   ├── database.yml        ← RDS PostgreSQL + Secrets Manager
│   ├── backend.yml         ← ECR, ECS cluster/service, ALB, IAM
│   ├── frontend.yml        ← S3 + CloudFront for the SPA
│   └─ lambda.yml           ← event-driven Lambda functions
├─ infra/lambda/            ← Lambda source
├─ backend/Dockerfile        ← API container image
└─ docker-compose.yml        ← local Postgres + API
```

14. Summary

SoleMate is deployed as a **containerized API on ECS Fargate** behind an ALB, with a **serverless event tier on Lambda**, a managed **RDS PostgreSQL** database, and the **React SPA served from S3 via CloudFront** — all defined in **CloudFormation** and shipped through an automated **GitHub Actions** pipeline. The design prioritizes scalability, security (private subnets, managed secrets, least-privilege IAM), zero-downtime deploys, and full observability, while keeping cost in check through right-sizing and serverless-for-bursty-work.